Consewative vector Fields

Determine whether F is a consewative vector field. If so, find a potential function for it.

1- $F(x,y) = xi + yj$

Sol)

$f(x,y) = x$, $g(x,y) = y$

Then,

$\frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x)$, $\frac{\partial g}{\partial x} = \frac{\partial}{\partial x}(y)$

$\frac{\partial f}{\partial y} = 0$, $\frac{\partial g}{\partial x} = 0$

So,

F is consewative.

Now,

$$\nabla \phi = F$$

$$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = xi + yj$$

comparing coefficients of i and j

$\frac{\partial \phi}{\partial x} = x \rightarrow (i)$, $\frac{\partial \phi}{\partial y} = y \rightarrow (ii)$

Integrate eq (i) wrt x and y wrt (ii)

$\int \frac{\partial \phi}{\partial x} dx = \int x\, dx$, $\int \frac{\partial \phi}{\partial y} = \int y\, dy$

$\phi = \frac{x^2}{2} + c$, $\phi = \frac{y^2}{2} + c$

Then,

$$\phi(x,y) = \frac{x^2}{2} + \frac{y^2}{2} + c$$

2) $F(x,y) = 3y^2 i + 6xy j$

Sol)

$f(x,y) = 3y^2$, $g(x,y) = 6xy$

$\frac{\partial f}{\partial y} = 6y$, $\frac{\partial g}{\partial x} = 6y$

So, F is conservative.

Then,

$\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = 3y^2 i + 6xy j$

comparing coefficients of i and j

$\frac{\partial\phi}{\partial x} = 3y^2 \to (i)$, $\frac{\partial\phi}{\partial y} = 6xy \to (ii)$

Integrate eq (i) wrt x and (ii) wrt y

$\int \frac{\partial\phi}{\partial x} dx = \int 3y^2 dx$, $\int \frac{\partial\phi}{\partial y} dy = \int 6xy\, dy$

$\phi = 3xy^2 + c$, $\phi = \frac{6xy^2}{2}$

$\phi = 3xy^2 + c$, $\phi = 3xy^2 + c$

$\phi(x,y) = 3xy^2 + c$

3) $F(x,y) = x^2 y i + 5xy^2 j$

Sol)

$f(x,y) = x^2 y$, $g(x,y) = 5xy^2$

$\frac{\partial f}{\partial y} = x^2$, $\frac{\partial g}{\partial x} = 10xy$

Not conservative

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4) $F(x,y) = e^x \cos y\, i - e^x \sin y\, j$

Sol)

$f(x,y) = e^x \cos y$, $g(x,y) = -e^x \sin y$

$\frac{\partial f}{\partial y} = -e^x \sin y$, $\frac{\partial g}{\partial x} = -e^x \sin y$

F is conservative

Now,

$\nabla \phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = e^x \cos y\, i - e^x \sin y\, j$

comparing coefficients of i and j

$\frac{\partial \phi}{\partial x} = e^x \cos y \rightarrow (i)$, $\frac{\partial \phi}{\partial y} = -e^x \sin y \rightarrow (ii)$

Integrate eq (i) wrt x and eq (ii) wrt y

$\int \frac{\partial \phi}{\partial x} dx = \int e^x \cos y\, dx$, $\int \frac{\partial \phi x}{\partial y} dy = -\int e^x \sin y\, dy$

$\phi = e^x \cos y$, $\phi = -e^x(-\cos y)$

$\phi = e^x \cos y + c$ $\quad$ $\phi = e^x \cos y + c$

$\phi(x,y) = e^x \cos y + c$

5) $F(x,y) = (\cos y + y\cos x) i + (\sin x - x\sin y) j$

Sol)

$f(x,y) = (\cos y + y\cos x)$, $g(x,y) = (\sin x - x\sin y)$

$\frac{\partial f}{\partial y} = -\sin y + \cos x$, $\frac{\partial g(x,y)}{\partial x} = \cos x - \sin y$

F is conservative

Now, $\nabla \phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = (\cos y + y\cos x) i + (\sin x - x\sin y) j$

comparing cofficients of i and j

$$\frac{\partial\phi}{\partial x} = \cos y + y\cos x \rightarrow (i)$$

$$\frac{\partial\phi}{\partial y} = \sin x - x\sin y \rightarrow (ii)$$

Integrate eq (i) w.r.t x and (ii) w.r.t y

$$\int\frac{\partial\phi}{\partial x}\,dx = \int(\cos y + y\cos x).\,dx$$

$$\phi = x\cos y + y\sin x + c$$

$$\int\frac{\partial\phi}{\partial y} \cdot dy = \int(\sin x - x\sin y)\,dy$$

$$\phi = y\sin x + x\cos y + c$$

$$\phi(x,y) = y\sin x + x\cos y + c$$

6) $F(x,y) = x\ln y\, i + y\ln x\, j$

Sol)

$f(x,y) = x\ln y$, $g(x,y) = y\ln x$

Now,

$$\frac{\partial f}{\partial y} = \frac{x}{y} \quad , \quad \frac{\partial g}{\partial x} = \frac{y}{x}$$

F is not conservative

Showthat the integral is independent of path, and find its value.

a) $\int_{(1,2)}^{(4,0)} 3y\,dx + 3x\,dy$

Sol)

$f(x,y) = 3y$, $g(x,y) = 3x$

$\frac{\partial f}{\partial y} = 3$, $\frac{\partial g}{\partial x} = 3$

F is conservative

Now,

$\nabla\phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = 3y i + 3x j$

comparing coefficients of i and j

$\frac{\partial \phi}{\partial x} = 3y \rightarrow (i)$, $\frac{\partial \phi}{\partial y} = 3x \rightarrow (ii)$

Integrate eq (i) w.r.t x and (ii) w.r.t y

$\int \frac{\partial \phi}{\partial x} dx = \int 3y\,dx$, $\int \frac{\partial \phi}{\partial y} dy = \int 3x\,dy$

$\phi = 3xy + c$, $\phi = 3xy + c$

$\phi(x,y) = 3xy + c$

Then,

$\int_{(1,2)}^{(4,0)} F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$

$\int_{(1,2)}^{(4,0)} 3y\,dx + 3x\,dy = \phi(4,0) - \phi(1,2)$

$= 0 - 6$

$= -6$ Ans:-

10) $\int_{(0,0)}^{(1,\frac{\pi}{2})} e^x \sin y \, dx + e^x \cos y \, dy$

Sol) $f(x,y) = e^x \sin y$, $g(x,y) = e^x \cos y$

$\frac{\partial f}{\partial y} = +e^x \cos y$, $\frac{\partial g}{\partial x} = +e^x \cos y$

F is conservative

Then,

$$\nabla \phi = F$$

$$\frac{\partial \phi}{\partial x} \cdot i + \frac{\partial \phi}{\partial y} j = e^x \sin y \, i + e^x \cos y \, j$$

comparing coffiicient of i and j

$\frac{\partial \phi}{\partial x} = e^x \sin y \rightarrow (i)$, $\frac{\partial \phi}{\partial y} = e^x \cos y \rightarrow (ii)$

Integrate eq(i) w.r.t x and (ii) w.r.t y

$\int \frac{\partial \phi}{\partial x} dx = \int e^x \sin y \, dx$, $\int \frac{\partial \phi}{\partial y} dy = \int e^x \cos y \, dy$

$\phi = e^x \sin y + c$, $\phi = e^x \sin y + c$

$$\phi(x,y) = e^x \sin + c$$

Now,

$$\int_{(0,0)}^{(1,\frac{\pi}{2})} F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$$

$$\int_{(0,0)}^{(1,\frac{\pi}{2})} e^x \sin y \, dx + e^x \cos y \, dy = \phi\left(1, \frac{\pi}{2}\right) - \phi(0,0)$$

$$\boxed{= e}$$

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11) $\int_{(0,0)}^{(3,2)} 2xe^y\,dx + x^2e^y\,dy$

Sol) $f(x,y) = 2xe^y$, $g(x,y) = x^2 e^y$

$\frac{\partial f}{\partial y} = 2xe^y$, $\frac{\partial g}{\partial x} = 2xe^y$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = 2xe^y i + x^2 e^y j$

comparing cofficients of i and j

$\frac{\partial \phi}{\partial x} = 2xe^y \rightarrow (i)$, $\frac{\partial \phi}{\partial y} = x^2 e^y \rightarrow (ii)$

Integrate eq (i) wrt x and eq (ii) wrt y

$\int \frac{\partial \phi}{\partial x} dx = \int 2xe^y\,dx$, $\int \frac{\partial \phi}{\partial y} dy = \int x^2 e^y\,dy$

$\phi = \frac{\cancel{2}x^2 e^y}{\cancel{2}}$, $\phi = x^2 e^y$

$\phi = x^2 e^y + c$, $\phi = x^2 e^y + c$

$\phi(x,y) = x^2 e^y + c$

Now,

$\int_{(0,0)}^{(3,2)} F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$

$\int_{(0,0)}^{(3,2)} 2xe^y dx + x^2 e^y dy = \phi(3,2) - \phi(0,0)$

$= 9e^2 - 0$

$= 9e^2$

Ans:-

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12) $\int_{(-1,2)}^{(0,1)} (3x-y+1)\,dx-(x+4y+2)\,dy$

Sol)

$f(x,y) = 3x-y+1 \quad , \quad g(x,y) = -(x+4y+2)$

$\frac{\partial f}{\partial y} = -1 \quad , \quad \frac{\partial g}{\partial x} = -1$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = (3x-y+1)i - (x+4y+2)j$

comparing coffiicients of i and j

$\frac{\partial\phi}{\partial x} = 3x-y+1 \rightarrow (i)$

$\frac{\partial\phi}{\partial y} = -x-4y-2 \rightarrow (ii)$

Integrate eq(i) wrt x and (ii) wrt y

$\int \frac{\partial\phi}{\partial x} dx = \int (3x-y+1)\,dx$

$\phi = \frac{3x^2}{2} - xy + x + c$

$\int \frac{\partial\phi}{\partial y} dy = \int (-x-4y-2)\,dy$

$\phi = -xy - \frac{4y^2}{2} - 2y$

$\phi = -xy - 2y^2 - 2y$

So, $\phi(x,y) = \frac{3x^2}{2} - xy + x - 2y^2 - 2y + c$

Then,

$\int_{(-1,2)}^{(0,1)} F\cdot dr = \phi(x_1,y_1) - \phi(x_0,y_0)$

$\int_{(-1,2)}^{(0,1)} (3x-y+1)\,dx - (x+4y+2)\,dy = \phi(0,1) - \phi(-1,2)$

$$= -2-2-\left(+\frac{3}{2}+2-1-8-4\right)$$

$$= -2-2-\frac{3}{2}-2+1+8+4$$

$$= \frac{11}{2} \text{ Ans:-}$$

13) $\int_{(2,-2)}^{(-1,0)} 2xy^3 dx + 3y^2x^2 dy$

Sol) $f(x,y) = 2xy^3$, $g(x,y) = 3y^2x^2$

$\frac{\partial f}{\partial y} = 6xy^2$, $\frac{\partial g}{\partial x} = 6xy^2$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = 2xy^3 i + 3y^2x^2 j$

comparing cofficients of i and j

$\frac{\partial\phi}{\partial x} = 2xy^3 \rightarrow (i)$, $\frac{\partial\phi}{\partial y} = 3y^2x^2 \rightarrow (ii)$

Integrate eq(i) wrt x and eq(ii) wrt y

$\int \frac{\partial\phi}{\partial x} dx = \int 2xy^3 dx$, $\int \frac{\partial\phi}{\partial y} dy = \int 3y^2x^2 dy$

$\phi = \frac{\cancel{2}x^2}{\cancel{2}} y^3$, $\phi = \frac{\cancel{3}y^3}{\cancel{3}} x^2$

$\phi = x^2y^3 + c$, $\phi = x^2y^3 + c$

$\phi(x,y) = x^2y^3 + c$

Now,

$\int_{(2,-2)}^{(-1,0)} F\cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$

$\int_{(2,-2)}^{(-1,0)} 2xy^3 dx + 3y^2x^2 dy = \phi(-1,0) - \phi(2,-2)$

$= 0 + 32$

$= 32$ Ans:-

14) $\int_{(1,1)}^{(3,3)} (e^x \ln y - \frac{e^y}{x}) dx + (\frac{e^x}{y} - e^y \ln x) dy$, where x and y are positives.

Sol)

$f(x,y) = e^x \ln y - \frac{e^y}{x}$, $g(x,y) = \frac{e^x}{y} - e^y \ln x$

$\frac{\partial f}{\partial y} = \frac{e^x}{y} - \frac{e^y}{x}$, $\frac{\partial g}{\partial x} = \frac{e^x}{y} - \frac{e^y}{x}$

F is conservative.

Then, $\nabla \phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = (e^x \ln y - \frac{e^y}{x}) i + (\frac{e^x}{y} - e^y \ln x)$

comparing cofficient of i and j

$\frac{\partial \phi}{\partial x} = e^x \ln y - \frac{e^y}{x} \rightarrow$ i)

$\frac{\partial \phi}{\partial y} = \frac{e^x}{y} - e^y \ln x \rightarrow$ ii)

Integrate eq (i) wrt x and eq (ii) wrt y

$\int \frac{\partial \phi}{\partial x} dx = \int (e^x \ln y - \frac{e^y}{x}) dx$

$\phi = e^x \ln y - e^y \ln x + c$

$\int \frac{\partial \phi}{\partial y} dy = \int (\frac{e^x}{y} - e^y \ln x) dy$

$\phi = e^x \ln y - e^y \ln x + c$

So, $\phi(x,y) = e^x \ln y - e^y \ln x$

Therefore,

$\int_{(1,1)}^{(3,3)} F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$

$\int_{(1,1)}^{(3,3)} (e^x \ln y - \frac{e^y}{x}) dx + (\frac{e^x}{y} - e^y \ln x) dy =$

$\phi(3,3) - \phi(\phi,\phi)$

$= \left(e^3 \ln 3 - e^3 \ln 3\right) - \left(e^0 \ln 0 - e^0 \ln 0\right)$

$= e^3 \cancel{\ln 3} - e^3 \cancel{\ln 3} - e^0 \cancel{\ln 0} + e^0 \cancel{\ln 0}$

$= 0$ Ans.-

Confiom that the force Field F is conservative in some open connected region containing the points P and Q, and then find the work done by the force field on a particle moving along an arbitoary smooth curve in the region from P to Q.

15) $F(x,y) = xy^2 i + x^2 y j$; $P(1,1)$, $Q(0,0)$

Sol)

$f(x,y) = xy^2$, $g(x,y)\ x^2y$

$\frac{\partial f}{\partial y} = 2xy$, $\frac{\partial f}{\partial x} = 2xy$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = xy^2 i + x^2 y j$

Comparing coefficients of i and j

$\frac{\partial\phi}{\partial x} = xy^2 \rightarrow (i)$, $\frac{\partial\phi}{\partial y} = x^2 y \rightarrow (ii)$

Integrate eq (i) w.r.t. x and eq (ii) w.r.t y

$\int \frac{\partial\phi}{\partial x} dx = \int xy^2 dx$, $\int \frac{\partial\phi}{\partial x} dy = \int x^2 y\, dy$

$\phi = \frac{x^2y^2}{2} + c$, $\phi = \frac{x^2y^2}{2} + c$

$\phi(x,y) = \frac{x^2y^2}{2} + c$

Now,

$\int_{(1,1)}^{(0,0)} F\cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$

$= \phi(0,0) - \phi(1,1)$

$= 0 - \frac{1}{2}$

$= \frac{-1}{2}$ Ans:-

16) $F(x,y) = 2xy^3 i + 3x^2y^2 j$; $P(-3,0), Q(4,1)$

Sol)

$f(x,y) = 2xy^3$, $g(x,y) = 3x^2y^2$

$\frac{\partial f}{\partial y} = 6xy^2$, $\frac{\partial g}{\partial x} = 6xy^2$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = 2xy^3 i + 3x^2y^2 j$

comparing coefficients of i and j

$\frac{\partial\phi}{\partial x} = 2xy^3 \to (i)$, $\frac{\partial\phi}{\partial y} = 3x^2y^2 \to (ii)$

Integrate eq (i) wrt x and (ii) wrt y

$\int \frac{\partial\phi}{\partial x} dx = \int 2xy^3 dx$, $\int \frac{\partial\phi}{\partial y} dy = \int 3x^2y^2 dy$

$\phi = \frac{\cancel{2}x^2}{\cancel{2}} y^3$, $\phi = \cancel{3} x^2 \frac{y^3}{\cancel{3}}$

$\phi(x,y) = x^2y^3 + c$

$$\int_{(-3,0)}^{(4,1)} F\cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$$

$$= \phi(4,1) - \phi(-3,0)$$

$$= 16 - 0$$

$$= 16 \text{ Ans:-}$$

17) $F(x,y) = ye^{xy} i + xe^{xy} j$; $P(-1,1), Q(2,0)$

Sol)

$f(x,y) = ye^{xy}$, $g(x,y) = xe^{xy}$

$\frac{\partial f}{\partial y} = e^{xy}$, $\frac{\partial g}{\partial x} = e^{xy}$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = ye^{xy} i + xe^{xy} j$

comparing cofficients of i and j

$\frac{\partial\phi}{\partial x} = ye^{xy} \rightarrow (i)$, $\frac{\partial\phi}{\partial y} = xe^{xy} \rightarrow (ii)$

Integrate eq(i) wrt x and (ii) wrt y

$\int \frac{\partial\phi}{\partial x} dx = \int ye^{xy} dx$, $\int \frac{\partial\phi}{\partial y} dy = \int xe^{xy} dy$

$\phi = \frac{\cancel{y}e^{xy}}{\cancel{y}}$, $\phi = \frac{\cancel{x}e^{xy}}{\cancel{x}}$

$\phi = e^{xy}$, $\phi = e^{xy}$

$\phi(x,y) = e^{xy} + c$

$$\int_{(-1,1)}^{(2,0)} F\cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$$

$$= \phi(2,0) - \phi(-1,1)$$

$$= 1 - e^{-1} \text{ Ans:-}$$

18) $F(x,y) = e^{-y}\cos x\, i - e^{-y}\sin x\, j$; $P(\frac{\pi}{2}, 1)$ $Q(\frac{-\pi}{2}, 0)$

Sol) $f(x,y) = e^{-y}\cos x$, $g(x,y) = -e^{-y}\sin x$

$\frac{\partial f}{\partial y} = -e^{-y}\cos x$, $\frac{\partial g}{\partial x} = -e^{-y}\cos x$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = (e^{-y}\cos x)i - (e^{-y}\sin x)j$

comparing coefficients of i & j

$\frac{\partial\phi}{\partial x} = e^{-y}\cos x \rightarrow (i)$, $\frac{\partial\phi}{\partial y} = -e^{-y}\sin x \rightarrow (ii)$

Integrate eq(i) wrt x and eq(ii) wrt y

$\int\frac{\partial\phi}{\partial x}dx = \int e^{-y}\cos x\, dx$, $\int\frac{\partial\phi}{\partial y}dy = -\int e^{-y}\sin x\, dy$

$\phi = e^{-y}\sin x$, $\phi = \cancel{-}\frac{e^{-y}\sin x}{\cancel{-1}}$

$\phi = e^{-y}\sin x + c$, $\phi = e^{-y}\sin x + c$

Therefore,

$$\phi(x,y) = e^{-y}\sin x + c$$

Now,

$\int_{(\frac{\pi}{2},1)}^{(\frac{-\pi}{2},0)} F\cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$

$= \phi(\frac{-\pi}{2}, 0) - \phi(\frac{\pi}{2}, 1)$

$= -1 - e^{-1}$

$= -1 - \frac{1}{e}$ Ans.

Find the exact value of $\int_C F \cdot dr$ using any method.

23- $F(x,y) = (e^y + ye^x) i + (xe^y + e^x) j$

$C: r(t) = \sin\left(\frac{\pi}{2} t\right) i + \ln t\, j \quad (1 \le t \le 2)$

Sol)

$f(x,y) = (e^y + ye^x) i$

$g(x,y) = (xe^y + e^x) j$

Now,

$\frac{\partial f}{\partial y} = e^y + e^x$

$\frac{\partial g}{\partial x} = e^y + e^x$

F is conservative

Then,

$\nabla \phi = F$

$\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j = (e^y + ye^x) i + (xe^y + e^x) j$

comparing cofficient of i and j

$\frac{\partial \phi}{\partial x} = (e^y + ye^x) \rightarrow (i)$

$\frac{\partial \phi}{\partial y} = (xe^y + e^x) \rightarrow (ii)$

Integrate (i) wrt x and (ii) wrt y

$\int \frac{\partial \phi}{\partial x} dx = \int (e^y + ye^x) dx$

$\phi = xe^y + y e^x + c$

$\int \frac{\partial \phi}{\partial y} dy = \int (xe^y + e^x) dy$

$\phi = xe^y + ye^x + c$

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Therefore,

$$\phi(x,y) = xe^y + ye^x + c$$

Now,

For $t=1$

$C: r(1) = \sin\left(\frac{\pi}{2}\right)i + \ln 1 j$

$= (1, 0)$

For $t = 2$

$C: r(2) = \sin\left(\frac{\pi 2}{2}\right) i + \ln 2 j$

$= (0, \ln 2)$

Now

$$\int_{(1,0)}^{(0,\ln 2)} F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$$

$$= \phi(0, \ln 2) - \phi(1, 0)$$

$$= \ln 2 - 1 \text{ Ans:-}$$

24) $F(x,y) = 2xy\, i + (x^2 + \cos y) j$

$C: r(t) = t i + t\cos\left(\frac{t}{3}\right) j \quad (0 \leq t \leq \pi)$

Sol)

$f(x,y) = 2xy \quad , \quad g(x,y) = (x^2 + \cos y)$

$\frac{\partial f}{\partial y} = 2x \quad , \quad \frac{\partial g}{\partial x} = 2x$

F is conservative

Then, $\nabla\phi = F$

$\frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j = 2xy\, i + (x^2 + \cos y) j$

comparing cofficients of i and j

$\frac{\partial\phi}{\partial x} = 2xy \rightarrow (i)$

$\frac{\partial\phi}{\partial y} = x^2 + \cos y \rightarrow (ii)$

Integrate eq (i) wrt x and eq (ii) wrt y

$\int \frac{\partial\phi}{\partial x} dx = \int 2xy\, dx$

$\phi = \frac{\cancel{2}x^2}{\cancel{2}} y$

$\phi = x^2 y + c$

$\int \frac{\partial\phi}{\partial y} dy = \int (x^2 + \cos y) dy$

$\phi = x^2 y + \sin y + c$

Thus,

$\phi(x,y) = x^2 y + \sin y + c$

Now,

For $t = 0$,

$C: \gamma(0) = 0 + 0 \cos(0) j$

$= (0, 0)$

For $t = \pi$

$C: \gamma(\pi) = \pi + \pi \cos\left(\frac{\pi}{3}\right)$

$$= \pi + \frac{\pi}{2}$$

$$= \left(\pi, \frac{\pi}{2}\right)$$

Now,

$$\int_{(0,0)}^{(\pi, \frac{\pi}{2})} F \cdot dr = \phi(x_1, y_1) - \phi(x_0, y_0)$$

$$= \phi\left(\pi, \frac{\pi}{2}\right) - \phi(0,0)$$

$$= \frac{\pi^3}{2} + 1$$ Ans:-